

System Impact Study Report

For

PJM Generation Interconnection Request
Queue Position AC1-109

Aurora 345kV

August 2020

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: (1) Direct Connections, which are new facilities and/or facilities upgrades needed to connect the generator to the PJM network, and (2) Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances a generator interconnection may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

The Queue AC1-109 project is an Aurora Generation, LLC (“Interconnection Customer” or “IC”), a LS Power owned facility, proposal to uprate the existing Aurora Generating Station, located at 2909 North Eola Road, Aurora, Dupage County, IL, 60502.

The AC1-109 project proposes a 30 MW increase in MFO and Capacity for Units 1 and 2.

Unit	Description	PSSE Bus No.	Existing CIRs	CIRs after AC1 Queue
1	GE PG7241FA	274737	152	152+15=167
2	GE PG7241FA	274739	152	152+15=167

These projects are all uprates of existing facilities and no electrical changes to the generators, transformers, etc. are proposed.

The IC has indicated the increased capability is available now. A backfeed date for this facility is not applicable.

This Generation Interconnection System Impact Study provides analysis results to aid the IC in assessing the practicality and cost of incorporating the facility into the PJM system.

Point of Interconnection

This is an existing facility and there will be no changes to the POI

Attachment Facilities

No new attachment Facilities

Direct Connection Network Upgrades

No new direct connection network upgrades are necessary

Network Impacts

The Queue Project AC1-109 was evaluated as a 30.0 MW (Capacity 30.0 MW) injection into the Aurora EC; RP 345 kV substation in the ComEd area. Project AC1-109 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AC1-109 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Analysis - 2020

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

Steady-State Voltage Requirements

(Results of the steady-state voltage studies should be inserted here)

None

Short Circuit

(Summary of impacted circuit breakers)

Not required

Affected System Analysis & Mitigation

MISO Impacts:

None

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request. Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

Not applicable

Light Load Analysis

Not required

Stability and Reactive Power Requirement

(Results of the dynamic studies should be inserted here)

No mitigations are required.

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)

None

Conclusion

Based upon the results of this System Impact Study, the increase of 30.0 MW (30.0 MW Capacity) to Units 1 and 2 of the Aurora Generating Facility (PJM Project #AC1-109) will not require additional interconnection charges.

Figure 1: Point of Interconnection Single-Line Diagram

AC1-109

